

Atmos ML Journal Club

August Posch 11 Sept 2026

Obs-to-obs weather
prediction, Zeus, and
Arraylake

Agenda

- Obs-to-obs forecasting – why do it, who is doing it
- How do we do it at Zeus
- Arraylake demo
- Discussion

Motivation

- Physical dynamics NWP (IFS, GFS, HRRR) first guess a current state in model space, then nudge it based on obs (“Data Assimilation”) to produce an analysis state (“initial conditions”) in model space, then forecast the next states.
 - Fancy MLWP models (GraphCast, GenCast, FourCastNet3) start from knowing an analysis state (“initial conditions”) in ERA5-space and then forecast the next states.
- We would like an MLWP that takes obs directly as inputs, and provides analysis and forecasts as outputs.

Obs-to-obs approach is gaining traction

- ECMWF is paying attention (AI-DOP)
- Nvidia is paying attention (HealDA)
- Google is paying attention (WeatherNext 3)
- Zeus has been here the whole time but we have a small budget

ECMWF – Pinnington et al. 2026

AIFS-DOP: END-TO-END MEDIUM-RANGE WEATHER PREDICTION FROM OBSERVATIONS ALONE WITH MACHINE LEARNING

 Ewan Pinnington  Peter Lean  Mihai Alexe  Eulalie Boucher  Simon Lang

 Patrick Laloyaux  Gert Mertes Tomas Kral

 Patricia de Rosnay

 Matthew Chantry  Anthony McNally

European Centre for Medium-Range Weather Forecasts (ECMWF)

June 18, 2026

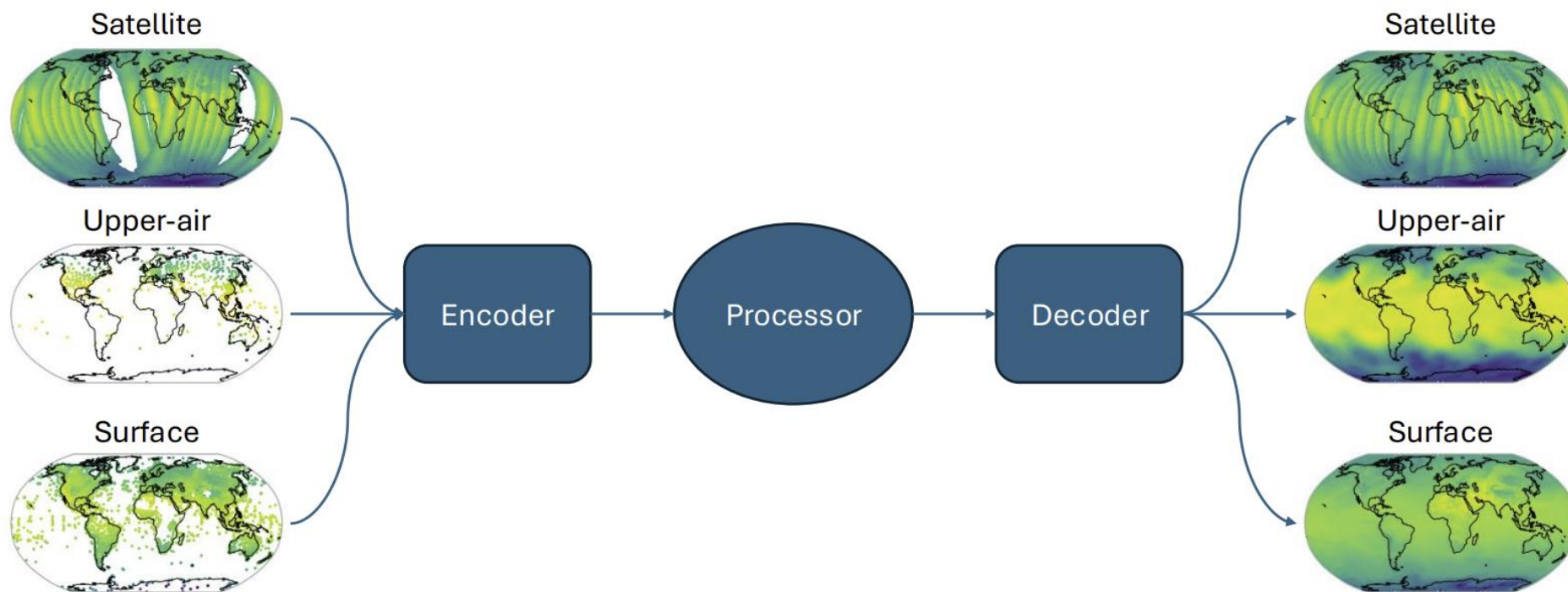


Figure 1: High-level model schematic: A single encoder is used for all observation types. The processor is as described in Lang et al. [2024] using a residual connection. Then a single decoder to predict observations out onto a full grid.

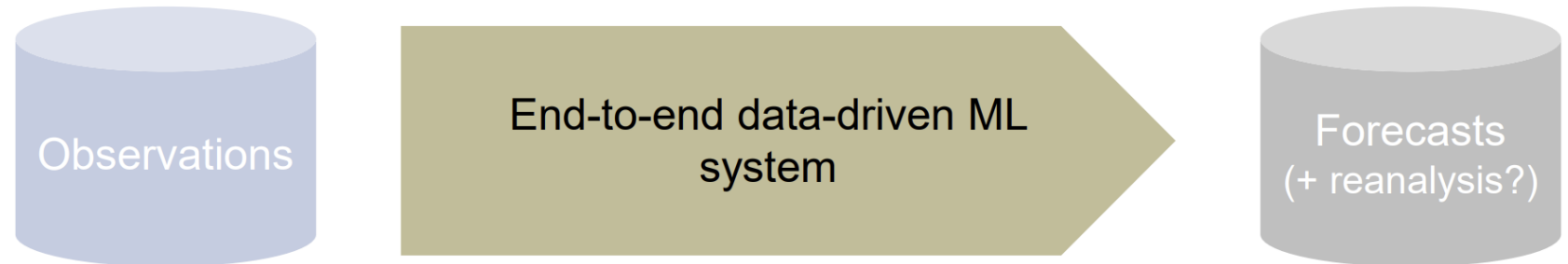
Context: Two parallel but complementary forecast streams

1) Physics/Hybrid (analysis-based)



→ Operational AIFS still has a hard dependency on 4D-Var (NEMOVAR / LDAS)

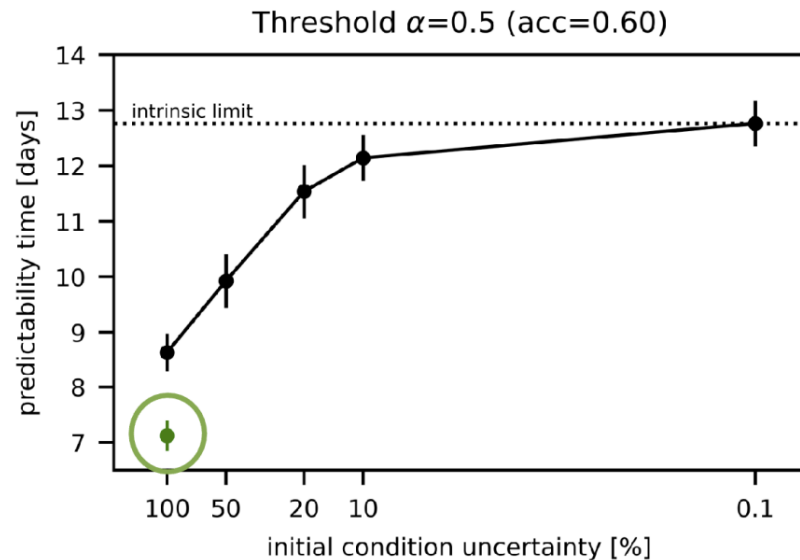
2) Pure ML (observation-based)



→ Ambition is to have a **pure ML observation-based end-to-end system**, what that system will look like is still to be determined – AI-DOP is one possibility, but research incoming from AIFS and WeatherGenerator as well

Motivation:

End-to-end learning from observations has potential to reduce initial condition errors



George Craig : ECMWF DA workshop 2025

- 1 to 2 days improvement possible from improved model
- 4 to 5 days improvement possible from improved initial conditions

Reanalysis-driven AI models are trained and initialized from the IFS initial conditions
→ learn to **reduce model errors**

Observation-driven AI models
→ can potentially **reduce initial condition *and* model errors**

The AI-DOP concept

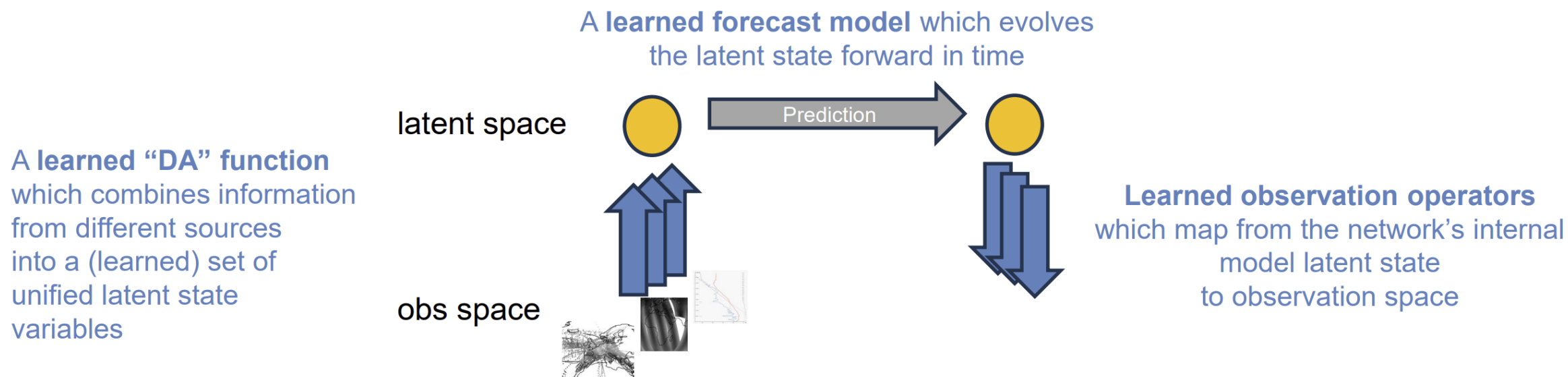
End-to-end observations to forecast

Training objective:

Using all satellite and in situ (conventional) observations in an input window

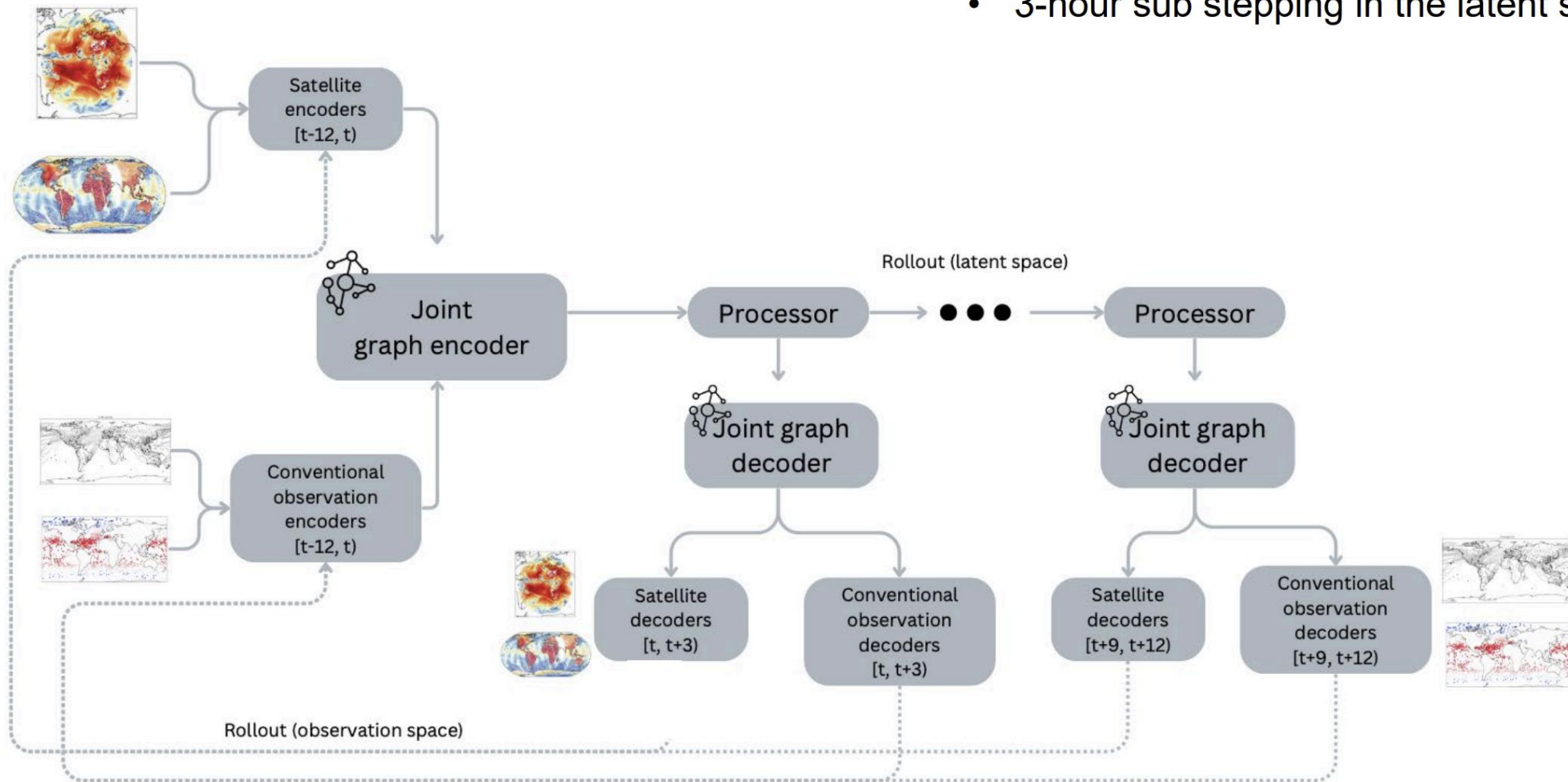
Forecast all observations in the next window **(with no reanalysis data used)**

To accurately predict future observations, the network must learn a representation of the state and processes that those observations are sensitive to



GraphDOP model architecture

- 10-20 year period for training
- 12-hour windows of observations
- 3-hour sub stepping in the latent space



Nvidia – Gupta et al. 2026

2026-1-27

HealDA: Highlighting the importance of initial errors in end-to-end AI weather forecasts

Aayush Gupta^{1,*}, Akshay Subramaniam¹, Michael S. Pritchard¹, Karthik Kashinath¹, Sergey Frolov², Kelsey Lieberman³, Christopher Miller³, Nicholas Silverman³ and Noah D. Brenowitz^{1,*}

¹NVIDIA Corporation, ²NOAA, ³MITRE Corporation, *Corresponding authors: Aayush Gupta (aaygupta@nvidia.com); Noah D. Brenowitz (nbrenowitz@nvidia.com)

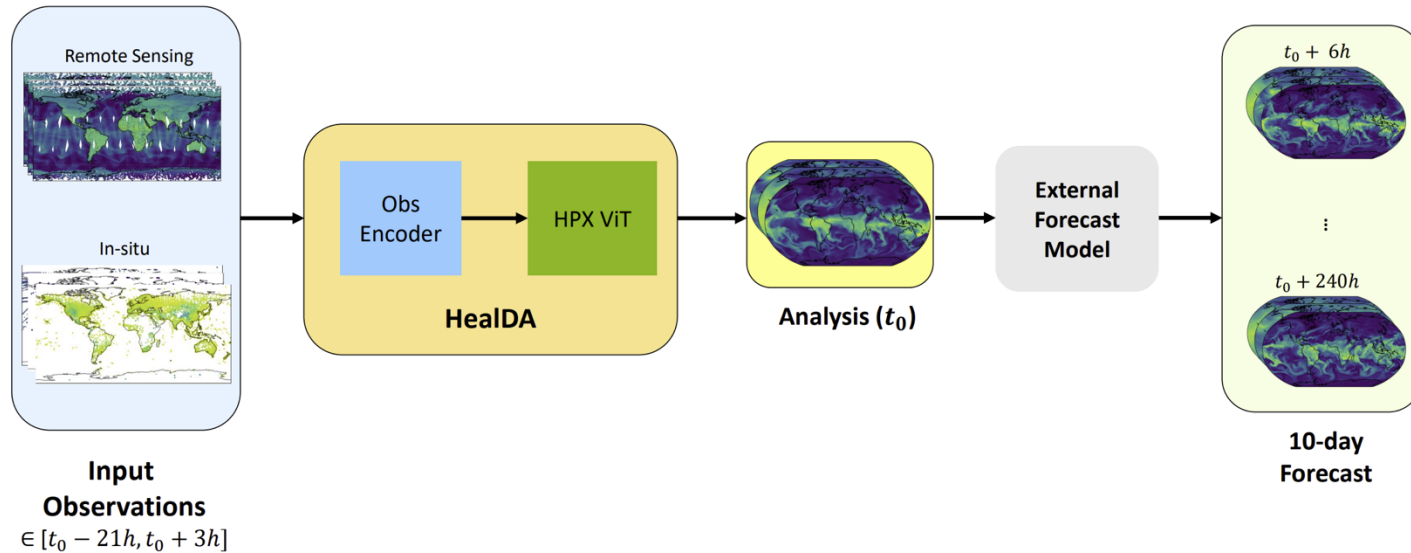


Figure 1: **End-to-end HealDA system and forecasting pipeline.** Observations from various remote-sensing instruments (ATMS, MHS, etc.) and in-situ sources (radiosondes, buoys, etc.) in the time window $[t_0 - 21\text{ h}, t_0 + 3\text{ h}]$ are processed by HealDA, which consists of an Observation Encoder (Obs Encoder) followed by an HPX ViT backbone, to produce an analysis state on the HPX grid at the target time t_0 . This analysis can then serve as the initial condition for an external forecast model (e.g., FCN3), to produce a 10-day forecast ($t_0 + 6\text{ h}, \dots, t_0 + 240\text{ h}$).

- HealDA converts raw observations into ERA5-like analysis state
- Allows GraphCast, GenCast, FCN3 to be used better operationally
- We ran HealDA with Zeus obs data feeds...

Google – Rasp et al. 2026

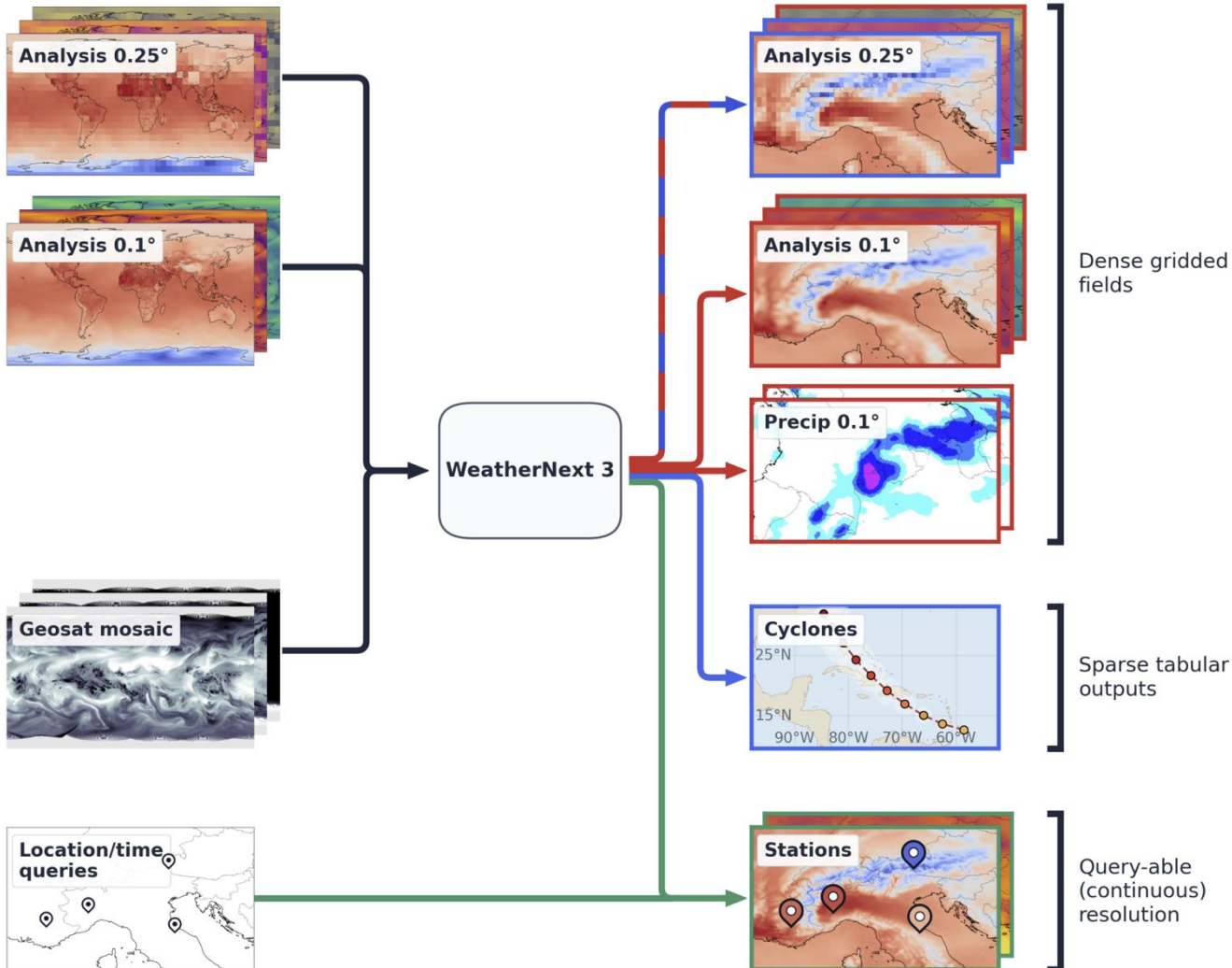
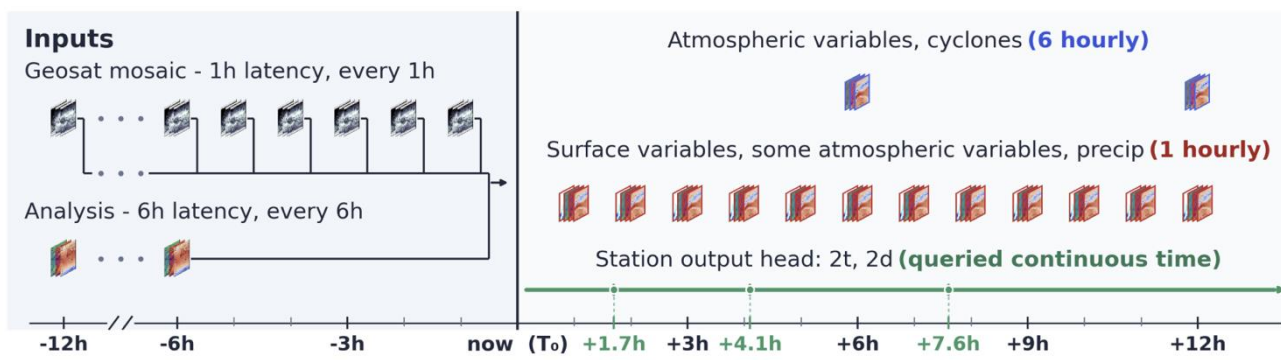
Google DeepMind

2026-9-4

WeatherNext 3: Increasing resolution and performance of global weather models with raw observations

Stephan Rasp^{1*}, Boris Babenko^{2*}, Dominic Masters^{1*}, Andrew El-Kadi^{1*}, Samier Merchant^{2*}, Guy Shalev^{1*}, Ilan Price^{1*}, Fred Zyda², Remi Lam¹, Sasha Shysheya¹, Matthew Willson¹, Stratis Markou¹, Shreya Agrawal¹, Suhani Vora¹, Mohammed Alewi Hassen¹, Sunny Mak¹, Tom R. Andersson¹, Megan Bela³, Akib Uddin², Nofar Peled Levi¹, Ben Gaiarin¹, Ferran Alet^{1*}, Aaron Bell², Peter Battaglia¹ and Alvaro Sanchez-Gonzalez^{1*}

*Equal contribution, ¹Google DeepMind, ²Google Research, ³Google

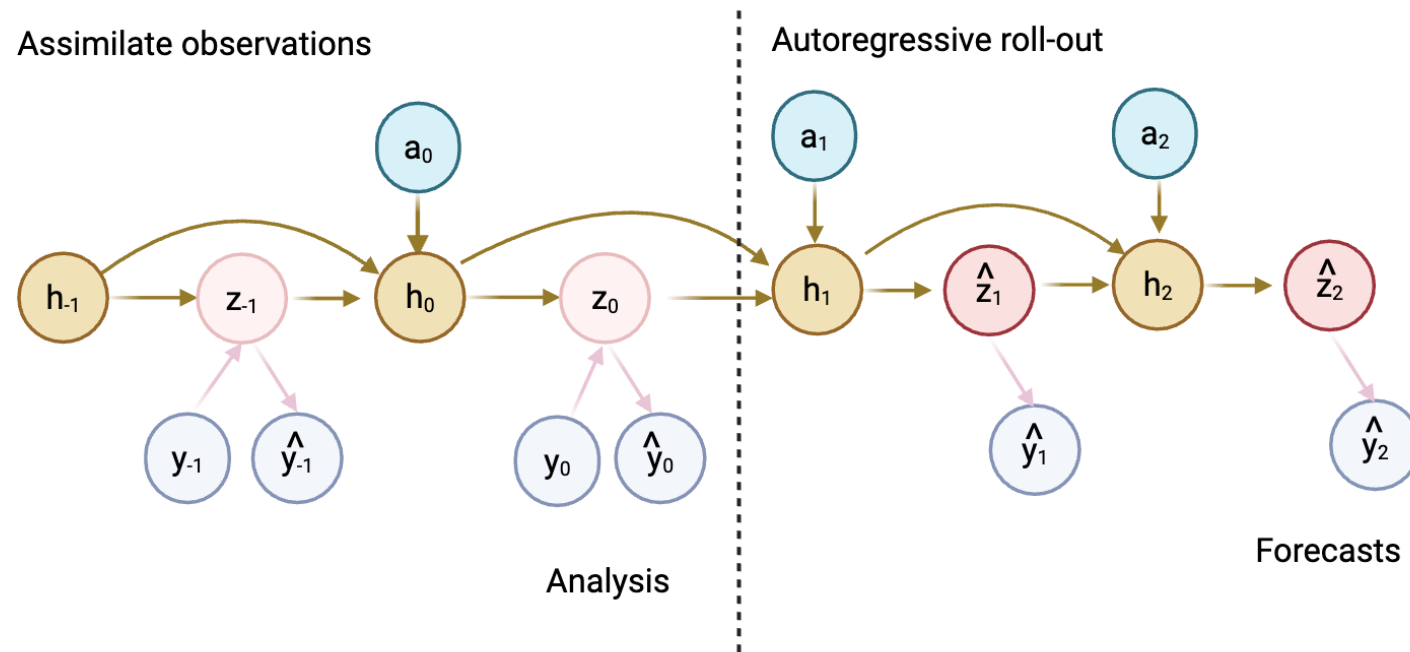


- WeatherNext 3 uses both kinds of inputs
 - physical-dynamics analysis state
 - low-latency observations from geostationary satellites

Why obs?

- Latency advantage
 - IFS analysis state (HRES) takes 5 hours to produce – if it's currently 05:00 and we want to predict what happens at 12:00, IFS/AIFS/FCN3 etc must use the 00:00 analysis state and make a 12-hour forecast.
 - Get the obs faster → gain lead time for same forecast.
- Hot take: Weather prediction used to be a physics problem. Now, it's a data engineering problem.

Obs-to-obs weather prediction with WeatherDreamer – Duffy et al. *in prep*



Temporal unrolling of the world model. The world model maintains a hidden state (h) that encodes a compressed history of all past observations and actions. A latent state (z) represents the current state estimate. During assimilation, the latent state is informed by the hidden state's prior information as well as new observations (x). During forecasting, generative dynamics model learns to perform next step prediction in the hidden space, such that the model can roll out trajectories as a Markov process, conditional on actions but without additional observational input.

Which obs?

- Brightness temperatures – Level 1B products
 - Contain a superset of Level 2 information (Level 2 means humidity, temperature, etc)
 - Available **fastest**
 - ML model can figure it out

Which obs? (currently live)

- Geostationary Infrared+Vis: ABI GOES-18/19, AMI GK2A, SEVIRI Meteosat
- Microwave Sounders: MHS Metop-B/C, AMSU-A Metop-B/C, ATMS NOAA-20/21/SNPP
- Infrared Sounders: CrIS NOAA-20/21
- Scatterometer: ASCAT Metop-B/C
- Radio Occultation: GRAS Metop-B/C, COSMIC-2, GRACE-FO, Sentinel-6A/B, GRAS-2 Metop-SG A1
- Not satellites: IGRA radiosondes, ACARS aircraft, ASOS stations

How do we get the obs

- Each sensor is different
- Example: Microwave Humidity Sounder on Metop-C (polar orbit)
 - Sensor does scans
 - Once per orbit, a bunch of scan files get dropped (dish)
 - Files processed/published (EUMETSAT data store)
 - We notice the scan files
 - We download, read the data, and put them on an hourly Healpix grid.
- We run WeatherDreameer on obs from all satellites to gap-fill and forecast and output different variables.

Latency!

How do we get the obs

- Each sensor is different
- Example: Microwave Humidity Sounder on Metop-C (polar orbit)
 - Sensor does scans
 - Once per orbit, a bunch of scan files get dropped (dish)
 - Files processed/published (EUMETSAT data store)
 - We notice the scan files
 - We download, read the data, and put them on an hourly Healpix grid.

0

5-105 mins

60-80 mins

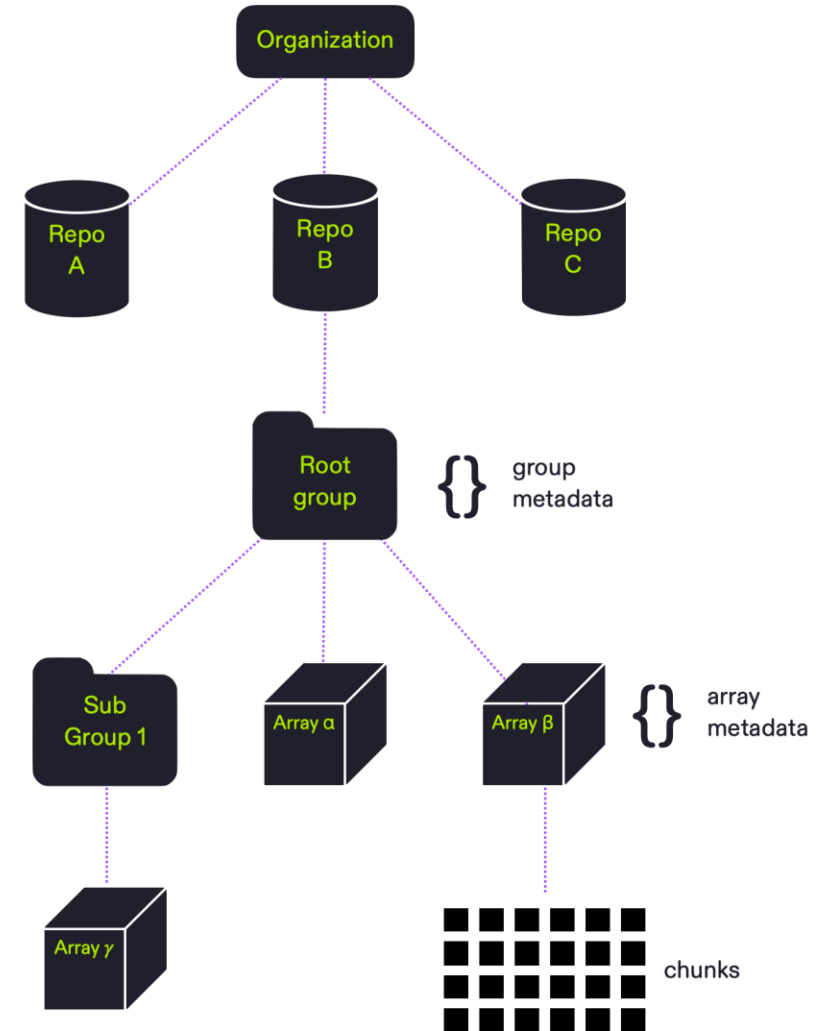
5-15 mins

5 mins

Total 2-3
hours. This
is one of the
slower ones.

Where to store the data

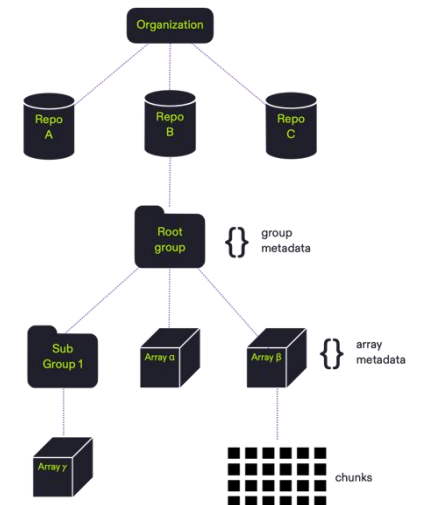
- Hourly healpix grid – how to store?
- We use Arraylake!



Where to store the data

- **Zarr** is a specification for storing large many-dimensional datasets
- **Icechunk** adds repositories for your zarrs, allowing version control, branches, commits, and database-like transactions.
- **Arraylake** provides a way to manage your icechunk repos in the cloud.

- Word bank: store, root, group, array, chunks



Demo

- Interactive guide to zarr: [An Interactive guide to Zarr](#)
- Click around Zeus repos – group structure, commits
- Arraylake Marketplace
- Start with GEO-IR notebook

Arraylake – start with GEO-IR

- Earthmover Marketplace dataset:
<https://app.earthmover.io/marketplace/6a0f68409ee79500fd510c3d>
- Colab notebook:
https://colab.research.google.com/drive/1ToN2i35epQSDGVf4Z1Pm57RvIkWnhizy?usp=sharing#scrollTo=F-jf-E_97WrK
- If you want to run locally, make a new python environment with **xarray** and **arraylake**!

ERA5 access

```
from arraylake import Client
import xarray as xr

client = Client()
repo = client.get_repo("earthmover-public/era5")
session = repo.readonly_session("main")

ds_spatial = xr.open_zarr(session.store,
group="single/spatial", chunks=None)
ds_spatial.t2m.sel(valid_time="2018-01-01
12:00").plot(cmap="coolwarm", robust=True)
```

Summary

- Obs-to-obs forecasting – why do it, who is doing it
- How do we do it at Zeus
- Arraylake demo
- Discussion

August Posch – OSB 320

always happy to meet for coffee

august@myzeus.ai ; aposch@uw.edu ; posch.au@northeastern.edu